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SHORT RESEARCH COMMUNICATION

Analysis of planetary and solar-induced perturbations on trans-Martian trajectory of Mars missions before and after Mars orbit insertion

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Abstract: Interplanetary missions are susceptible to gravitational and nongravitational perturbing forces at every trajectory phase, assuming, of course, that the man made rockets and thrusters work as expected. These forces are mainly due to planetary and solar-forcing-induced perturbations during geocentric, heliocentric and Martian trajectories, and before orbit insertion. In this study, we review and/or analyze Mars orbiters mission associated perturbing forces and their possible impacts before Mars Orbit Insertion viz Earth's oblateness, Third body (solar and lunar), solar radiation pressure, solar energetic radiation environment and atmospheric drag forces. We also model the significance of atmospheric drag force on Mangalyaan Mars orbiter mission, as a function of appropriate space environmental parameters during its 28 days in Earth's orbit (around and during perigee passage), 300 days of heliocentric and 100 days of Martian trajectory. We have found that for a total perigee height boost of about 250 km, the cumulative orbit decay can be approximately 720 m. The approximate altitude variation could be up to 158 m with respect to the sun during 300 days of interplanetary journey toward Mars. After Mars orbit insertion, the total decay experienced by the spacecraft could be up to 701 m with decay rate of up to 9 m/day during 100 days of Martian trajectory, based on Mars–Earth atmosphere density ratio. In principle, resulting deviations due to perturbing forces are usually corrected before Earth departure (and/or Mars orbit insertion) and are beyond the scope of this work. However, the knowledge is important for mission planning, design, implementation and situational awareness. We find that the deviations are small enough and should be correctable.

Keywords: Dynamics of atmosphere; Interplanetary physics; Satellites; Space weather

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1. Introduction

Driven by the quest for the possibility of life in Mars, clues to the evolution of our solar system, fascination with the chemistry, geology and meteorology of another planets, the National Aeronautics and Space Administration (NASA) has launched the first (successful) Mars mission (Mars orbiter and lander—Viking 1) in August 1975, which has arrived near Mars on the June 19 and has landed on July 20, 1976 [1]. While Viking 1 mission has operated between June 1975 and 1980, more than four other orbiter and/or lander missions have also made it to the 'red planet' within about three decades. Current active Mars missions include Mars Odyssey (NASA/USA, launched in April 2001 with arrival in October 2001 [2]), Mars Express and Beagle 2 (ESA/Europe, launched in June 2003 with arrival in December 2003 [3]), Mars Exploration Rover Opportunity (NASA/USA, launched in July 2003 with arrival in January 2004 [4]), Mars Reconnaissance Orbiter (NASA/USA, launched in August 2005 with arrival in March 2006 [5]) and Curiosity Mars Science Laboratory (NASA/USA, launched in November 2011 with arrival in August 2012). Russia, the Soviet Union, had also recorded a 'short-lived' success in their past attempts with MARS 2, 3, 5 and 6 missions (among others). The Indian Space Research Organisation (ISRO) in her first ever attempt has launched an interplanetary (Mars) mission on November 5, 2013. The country's Mars orbiter mission (also known as

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Mangalyaan) orbited the Earth between November 5, 2013 and December 1, 2013, basically building up the required velocity ($+\Delta V$), which is needed to escape the Earth's sphere of influence (ESOI). It has begun a 300-day journey to Mars on December 1, 2013 with successful Mars orbital insertion on September 24, 2014.

Quite a large number of attempts have been made in the past, toward interplanetary missions (IPMs), but only a few successes have been recorded so far. It has been estimated that more than half of the attempted missions to explore the 'red planet' (Mars) failed [6]. However, instructive and resourceful lessons have been learned from each failure, which has made subsequent attempts more successful [1, 7]. Although IPM attempts date back to the 1960s, more than half of successful ones have been launched within the last two decades. The high failure rate in IPM may be associated with planetary and solar-induced perturbations (after launch) among other causality, especially during transit and arrival to their destined planet. High precision and accuracy in calculations are important to successful design and implementation of interplanetary missions. Very small and apparently insignificant parametric fluctuations (in some cases) due to perturbing forces may introduce sporadic errors that can hamper the success of such IPMs. Other potential causes of IPMs failure include under- and/or over-performance of thrusters, miscalculations in firing directions and Liquid Apogee Motor (LAM) alignments, failure of launch vehicles, communication/radio failures and nature/mode of entry [6–9]. Mangalyaan is not an exception to these IPM-associated challenges.

This work is a pointer to some sources of fluctuations that may arise due to planetary and solar-forcing-induced perturbations. We review some IPM-associated perturbing forces with reference to Mangalyaan spacecraft, based on established theory. In detail, we also analyze (by model) and investigate the significance of atmospheric drag force on Mangalyaan spacecraft, as a function of space environmental parameters during geocentric (28 days in Earth's orbit), heliocentric (300 days in the sun's orbit) and Martian trajectory (assumed to be up to 100 days in Mars orbit (after MOI), in this paper for concreteness). Atmospheric drag causes change in orbital parameters making it difficult to identify and track satellites and other space objects, maneuvering and predicting lifetime and reentry [10–13]. In principle, deviations due to associated perturbations under consideration (e.g., Decay, Solar Radiation Pressure, Third body) are corrected before Earth departure and/or MOI. Such adjustments are beyond the scope of this work. However, the knowledge of their relative significance is important to mission planning, design, implementation and situational awareness.

2. Interplanetary trajectories and Mars mission system engineering challenges

Interplanetary mission requires a succession of transfer of a satellite from one orbit to another by means of a change of velocity (ΔV) as the system moves through successive phase mission plan to a destined planet. For a Mars-bound spacecraft (from Earth), three-phase trajectory is required—geocentric, heliocentric and Martian. In this paper, our emphasis will be on Mars interplanetary missions. One common type of orbit transfers employed in interplanetary missions is the Hohmann transfer or trajectory. The Hohmann transfer is considered to be the minimum two-impulse transfer between coplanar circular orbits [14, 15]. This type of transfer requires elliptical paths that are tangent to the launch and arrival orbits. Three main groups of trajectories can evolve from the Hohmann transfer—staying tangential to the larger orbit but intersecting the smaller one, intersecting the larger orbit and staying tangential to the smaller one and intersecting both orbits [15]. The equation of Earth departure, heliocentric and Mars arrival trajectory velocities of a Mars interplanetary mission is given by the following:

(i) Earth departure velocity

$$\upsilon_{\rm bo} = \sqrt{2\left[\frac{\mu_{\rm e}}{R_{\rm e}+h} + E_{\rm e}\right]} \tag{1}$$

where $\upsilon_{\rm bo}$ is the burnout velocity, $R_{\rm e}$ is the equatorial radius of Earth, h is the altitude at injection, $\mu_{\rm e}$ is the gravitational parameter of the Earth and $E_{\rm e}$ is the energy of escape hyperbola.

(ii) Heliocentric trajectory velocity at perihelion is

$$\upsilon_{\rm p} = \sqrt{\mu_{\rm s}\left(\frac{2}{r_{\rm e}} - \frac{1}{a}\right)} \tag{2}$$

The transfer orbit at apoapsis is

$$\upsilon_{\rm a} = \sqrt{\mu_{\rm s}\left(\frac{2}{r_{\rm m}} - \frac{1}{a}\right)} \tag{3}$$

The velocity at exit from the Earth's sphere of influence is

$$\upsilon_{\infty} = \upsilon_{\rm p} - \upsilon_{\rm e} \tag{4}$$

where $r_{\rm m}$ is the radius of Mars, $r_{\rm e}$ is the radius of the Earth at 1 A.U., a is the apoapsis $(r_{\rm e} + r_{\rm m}/2) = 1.262$ A.U., $\mu_{\rm s}$ is the gravitational parameter of the sun, υ_e is the mean velocity in Earth's orbit.

(iii) Mars arrival phase: The retro velocity at Mars surface is

$$\upsilon_{\text{retro}} = \sqrt{2\left(\frac{\mu_{\text{m}}}{R_{\text{m}}} + E_{\text{m}}\right)} \tag{5}$$

where μ_{m} is the gravitational parameter of Mars, R_{m} is the equatorial radius of Mars and E_{m} is the energy of the hyperbolic orbit at Mars [15]. E_{m} is given by Eq. (6):

$$E_{\text{m}} = \frac{\upsilon_{\infty}^2}{2} = \frac{(\upsilon_{\text{m}} - \upsilon_{\text{a}})}{2} \tag{6}$$

Depending on the type of mission (orbiter or lander), Mars orbiter spacecrafts begin their missions after successful orbit insertion, mainly exploration of Mars surface and/or atmosphere. Mars lander missions (e.g., Curiosity Rover) require entry, descent and subsequent landing in the red planet. This is yet another challenging phase. Braun and Manning [16] in their study has pointed out some system challenges associated with Mars exploration entry, descent, and landing emanating from three sources: (i) an atmosphere which is thick enough to create substantial heating, but not low enough to reduce terminal descent velocity, (ii) a surface environment of complex rocks, craters, dusts and terrain patterns and (iii) the cost of replicating a Mars-relevant environment for space flight qualification of new entry, descent and landing technologies [16]. In our present situation, we would be interested in shrinking of the orbits due to repeated passage of Mangalyaan at the perigee, interplanetary phase and periapsis once in the Marian orbit.

3. Mangalyaan Mars orbiter mission at a glance

The Mangalyaan MOM is launched aboard Polar Satellite Launcher Vehicle PSLV-XL. At delivery, the observed initial perigee height is about 248.4 km, apogee 23,500 km and inclination = 19.2°, where it has used its own propulsion system to insert itself into its trans-Martian interplanetary trajectory within a period of about 4 weeks. The space probe has a mass of about 1337 kg (with a dry mass of about 475 kg, including five payloads of about 15 kg) and carries a fuel of about 852 kg. It has similar core structure and spacecraft system and mission largely based on the Chandrayaan-1 Moon Orbiter. Mangalyaan is equipped with a single deployable solar array that consists of three panels (each being 1.4 × 1.8 m in size) with yoke and drive mechanism, capable of providing up to 840 Watts of electrical power at Mars. It is fed to a power distribution unit that provides power to the various systems and payloads and also controls the state of charge of a 36-amp-h battery for night passes. Other compositions of the MOM include bi-propellant main propulsion system and an altitude control system, four reaction wheels, a 2.2-m-diameter High Gain Antenna and a number of other High Technological equipment that meets critical mission operations and stringent requirements. The 15-kg payload consists of a suite of five science instruments that equip MOM for its distinct mission objectives—Lyman Alpha Photometer (LAP), Martian Exospheric Neutral Composition Analyzer (MENCA), Mars Colour Camera (MCC), Methane Sensor for Mars (MSM) and Thermal Infrared Imaging System (TIS) (also see [17, 18]). Mangalyaan is saddled with the mission to explore Mars surface feature, morphology, topography, mineralogy and Martian atmosphere. On a specific mission, it will carry out the search for methane on the 'red planet.' The pictorial representation of the deployed and dissembled view of Mangalyaan is shown in Fig. 1(a) and 1(b).

4. Mangalyaan mission plan and the trans-Martian flight profile

After delivery to an initial elliptical orbit of 248.4 km (perigee, r_{p}) by 23,500 km (apogee, r_{a}) and inclination of about 19.2°, the Mars probe has to enter three phases of mission plan to reach Mars—(i) the geocentric phase, (ii) the heliocentric phase and (iii) the Mars-centric (areocentric) phase. The three phases required that MOM fires its liquid apogee motor (LAM) six times at a given interval when passing perigee, to gradually increase the apogee of the orbit and consequently moves through the phases up until departure to Mars. The trajectory design is shown in Fig. 2(a). At the geocentric phase, the spacecraft with six engine burns gradually maneuver into a 'depart' hyperbolic trajectory with which it escapes from the Earth's sphere of influence and with orbital velocity boost. Beyond the Earth's sphere of influence, the perturbing force on the orbiter is due to the sun. Between November 5, 2013 and November 16, 2013, the apogee has been raised to approximately 192,874 km from the initial 23,500 km after five consecutive raising maneuvers. In the computation which follows, we have assumed a corresponding perigee rises to approximately 500 km from an initial of about 248.4 km (a total incremental height of 250 km) during velocity boost ($+\Delta V$). The maneuvered apogee heights after each velocity boost are shown in Fig. 2(b).

This work has been started shortly after the launch of the spacecraft with continuous update until MOI. It has departed Earth's orbit and tangentially (to its orbit) encountered Mars orbit, as shown in Fig. 2(a). At the areocentric orbital phase, the spacecraft has reached the Mars sphere of influence in a hyperbolic trajectory on September 24, 2014. At closest approach to Mars, it is captured into the planned orbit around Mars by imparting ΔV retro (also Mars orbit

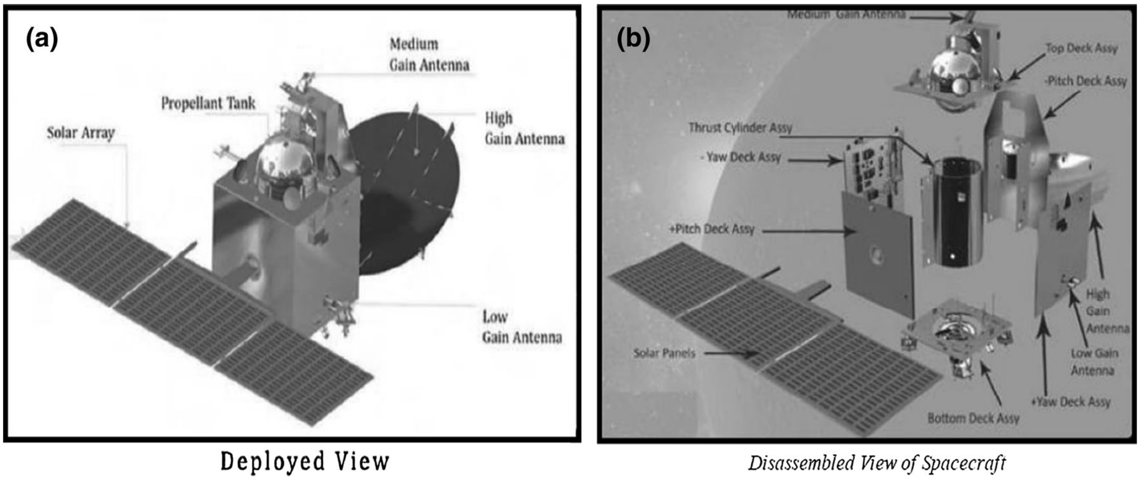


Fig. 1 (**a**) Deployed view of Mangalyaan spacecraft, (**b**) the dissembled view [17]

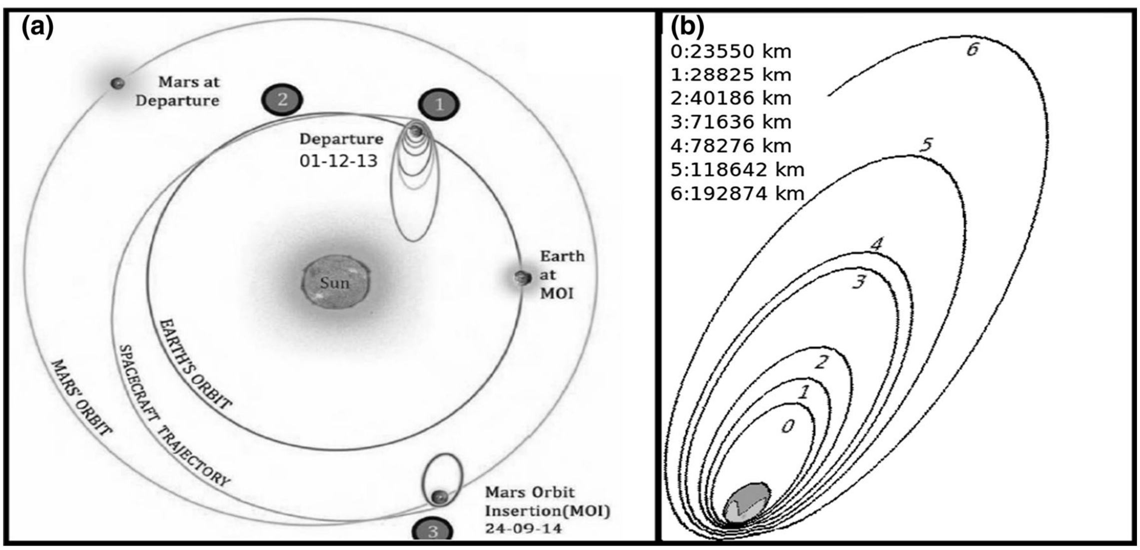


Fig. 2 (**a**) Planned trans-Martian trajectory of Mangalyaan spacecraft, (**b**) maneuvered orbits (heights) after consecutive velocity boost (adapted from [17])

Insertion) maneuver [17]. Figure 3 shows planned 'Mangalyaan' trajectory in Martian orbit.

5. Procedure

In this work, we have analyzed IPM-associated perturbing forces and their possible impact on trans-Martian trajectory and/or mission (before MOI), based on established theory. In Sect. 6.4, we have computed orbital decay of 'Mangalyaan' due to atmospheric drag during geocentric, heliocentric and Martian phase trajectories. We have assumed that drag effect around and during apogee passage is negligible during geocentric trajectory because of the large distance between the spacecraft and the Earth. We have computed drag force impact on the spacecraft as a function of thermospheric density and space environmental parameters around and during the perigee passage for 28 days in Earth's orbit, 300 days heliocentric trajectory and 100 days of Martian orbit trajectory. The elliptic orbit geometric analysis of the spacecraft in Earth and Martian orbit at perigee distances for atmospheric density and drag analysis is given in Sect. 6.4.3.

5.1. Mangalyaan orbital parameters required for our study

The effective exposed area of the spacecraft (in the direction of motion) is critical to atmospheric drag force. We have estimated this parameter from the satellite's mainframe elements and/or specifications. Hence, we have considered three parts of relative importance—the solar

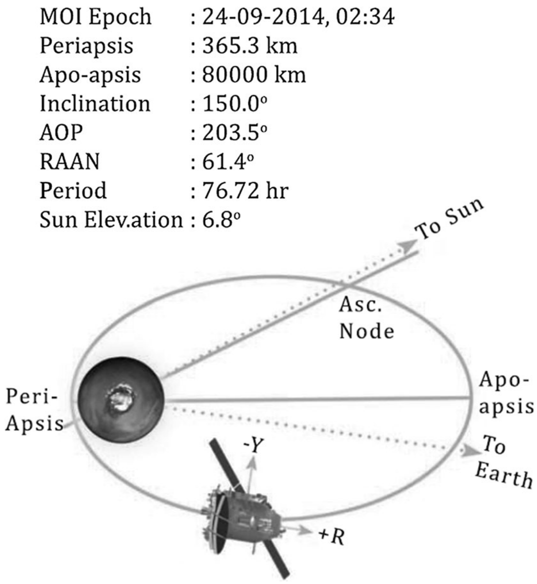


Fig. 3 Planned orbital parameters and trajectory of Mangalyaan spacecraft

panels, the Yaw and Pitch deck assembly and the top and bottom deck assembly. The dissembled view of the spacecraft is shown in Fig. 1b, indicating the mainframe view. For simplicity, each of the deck assembly is assumed to be square-shaped of approximately 3.34 m^2. It is generally the case that the solar panel wings of most satellites are usually designed to constantly adjust to maintain an optimum amount of battery charging and minimize frontal area projected to the ram direction, thereby minimizing the drag (which also saves propellant) [19]. We, therefore, have assumed the projected surface area (during its trajectory) to be about 5.04 m^2. This value may change due to solar panel directional changes, which are usually offset with respect to the sun direction [20].

6. Analysis of perturbations associated with mangalyaan MOM trajectory

Space probes must be weather-strong perturbing forces during their mission in space to survive. The trans-Martian trajectory and flight profile of IPM has been explained in Sects. 2 and 4. In general, IPMs experience gravitational perturbing force, mainly solar/lunar gravitational attractions and Earth's oblateness (J_2) and its triaxiality and nongravitational force such as atmospheric drag, solar radiation pressure and/or environment, outgassing and tidal effects [15]. The forces that influence the trajectory of the mission vary from one phase of the trajectory to another, and the probability and/or extent of these perturbation forces driven impacts may also be affected by factors such as phase of the 11-year solar cycle, nature of the spacecraft orbit, local time and position of the satellite relative to Earth–Sun direction [21]. The induced variations (by perturbing forces) on the orbital elements of the system may be secular, short-period or long-period, or a combination of such variations. Secular variations constitute a linear variation in the element. Short-period variations represent periodic variations with a period less than or equal to the orbital period and long-period variations with a period greater than the orbital period [9]. We have analyzed some of the associated orbit perturbing forces on the orbit of MOM during geocentric, heliocentric and Martian trajectories. The general form of motion and perturbations associated with satellite trajectory is given by Eq. (7):

$$\frac{d^2 r}{dt^2} = -\frac{\mu}{r^3}\bar{r} + a_p \tag{7}$$

where a_p is the sum of acceleration caused by perturbation forces. The perturbing acceleration and/or force may be gravitational or nongravitational (as mentioned earlier).

In general, the perturbing acceleration a_p of a satellite due to a perturbing body having a mass M_p and gravitational parameter is given by the equation:

$$a_p = \mu_p \sqrt{(R.R)} \tag{8}$$

where

$$R = \frac{r_{sp}}{r_{sp}^3} - \frac{r_p}{r_p^3}$$

where r_{sp} is the distance between the satellite and the 'perturber' and r_p is the distance of the perturber from the planet. Expressions that provide approximate average rates of change of orbital elements for a single disturbing body can be found in [22, 23]. The relative magnitudes of some of the sources of perturbations acting upon an Earth-orbiting spacecraft are illustrated in Fig. 4. For each effect, the logarithm of the disturbing acceleration, normalized to 1 g, is shown as a function of altitude. Clearly, the effect of the drag force of the earth becomes very week near the apogee of the earth orbit, but important near the perigee.

6.1. Earth's oblateness (J_2) effects

Mangalyaan, like any other interplanetary mission, may experience effects of Earths oblateness (J_2) during the first 28 days, especially at the altitude (height) around the perigee [9, 15, 22, 23]. This is mainly an off-center gravitational pull due to Earth's equatorial bulge. The principal effects of the J_2 zonal harmonic or Earth oblateness are secular motions of the node (Ω) and perigee (ω) of an orbit. This introduces a force component toward the equator. The resultant acceleration causes the satellite to reach the equator (node) short of the crossing point for a spherical Earth [15]. This effect becomes less important

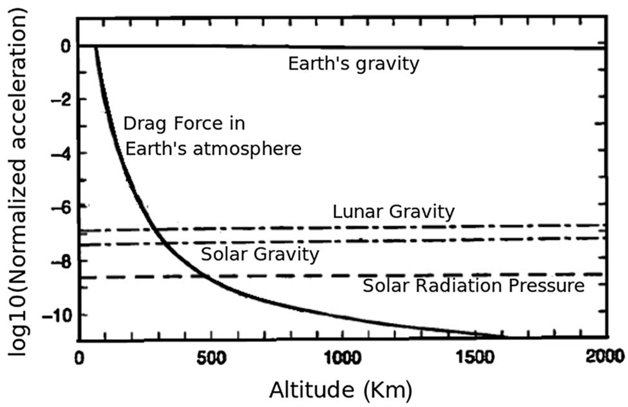


Fig. 4 Relative magnitudes of main sources of perturbation acting upon an Earth-orbiting spacecraft (Adapted from [9])

with increasing perigee distance from the Earth, at which solar and lunar gravitational perturbations become significant, especially at or around the apogee height. The only effect to compete with these gravity-induced effects is aerodynamic drag [9]. The effects of oblateness on the right ascension of the ascending node, argument of pericentron and mean anomaly were calculated from the prescriptions given in [23, 24].

6.2. Third-body perturbations

There is a significant gravitational pull from the sun and moon (solar and lunar) especially around and during the apogee passage, which increases as it builds up both velocity and altitude. The gravitational forces of the sun and moon (and other bodies in the solar system) cause periodic variations of most of the orbital elements. The right ascension of the ascending node, argument of perigee and mean anomaly experiences secular variations [9, 22, 23, 25]. These secular variations arise from a 'spinning' (gyroscopic) precession of the orbit about the ecliptic pole. The relative magnitude of solar and lunar perturbations acting upon an Earth-orbiting spacecraft is shown in Fig. 4.

6.3. Solar radiation pressure (SRP)

SRP on satellites is due to the impingement and absorption, and reflection of photons from the sun on the surface of a satellite including on the solar panels. The main effect of this phenomenon is a change in the eccentricity and longitude of perigee. The extent of variation of these elements depends on the effective area, surface reflection and inverse of satellite's mass [15, 26]. The effects caused by solar radiation pressure exceed that of atmospheric drag at altitudes above 900 km, as shown in Fig. 4. The changes in perigee height induced by SRP can have significant effects on satellite lifetime [15]. The perturbing acceleration of a satellite due to SRP-induced effects can be computed from Eq. (9) [26]:

$$\vec{a}_{\mathrm{srp}} = -\rho_{\mathrm{sr}} \frac{C_{\mathrm{r}} A_{\mathrm{sun}}}{m} \frac{\vec{r}_{\mathrm{sat-sun}}}{\vec{r}_{\mathrm{sat-sun}}} \tag{9}$$

where ρ_{sr} is the incoming pressure which depends on the time of the year and the intensity of the solar output. It is derived from the incoming solar flux and values of about 1358–1373 W/m^2. C_{r} is the coefficient of reflectivity. It depends on the absorptive properties of the material and thus the susceptibility to incoming solar radiation. A_{sun} is the cross-sectional area, which changes constantly (without altitude control). It can vary by up to a factor of 10 or more depending on satellite configuration. m is the mass of the satellite. Although m is usually constant but can be influenced by factors such as thrusting and ablation. $r_{\mathrm{sat-sun}}$ is the orientation of the satellite-sun vector. Detail equations and derivation of variations in Keplerian parameters can be found in [27–29]. We have, therefore, not repeated the derivations here.

6.4. Atmospheric drag force effect

The Mangalyaan is susceptible to significant atmospheric drag force around and during the apogee passage in Earth orbit (geocentric orbit). This effect is insignificant in heliocentric orbit due to the large distance of the spacecraft from the sun and its transit origin (the Earth). In Martian orbit (and atmosphere), drag effect is important but much less compared to that in Earth orbit (which, of course, depends on the type of orbit), especially on a long-term trajectory profile. Mangalyaan would experience significant drag around the periapsis (365.3 km) over time in its areocentric orbit. We, therefore, investigate the significance of these effects at each trajectory phase.

Atmospheric drag on low Earth-orbiting satellites (>1000 km) is primarily caused by solar-forcing-induced variations in thermospheric density profile. Energetic particles (and EM radiations) emitted from the sun during solar energetic events (e.g., solar wind streams, coronal mass ejections, solar flares) are subsequently deposited in the upper atmosphere [30, 31]. The upper atmospheres heat up and expand as a consequence and alter (change) thermosphere density profile, leading to accelerated drag on satellites [31–33]. Atmospheric drag, therefore, varies in direct proportion with atmospheric density. The prediction of the lifetime, reentry and/or computation of drag on low Earth orbit satellites (LEOSs) largely depend on a good knowledge of the variations of thermospheric densities, which is an important space environmental parameter for satellite operations in near-Earth space [34, 35]. Although this quantity is not precisely known at any given instant, but many empirical atmospheric models have been developed and more are being developed, with increasing sophistication and good approximation [36–40]. We use one such model, the *NRLMSISE-00* empirical atmospheric model, in our drag model for a good representation of the space weather condition under which this space system traverses. Atmospheric drag experienced by a satellite at perigee can significantly lower the apogee, causing the orbit to become more and more circular, until the entire orbit is at the perigee altitude. This situation can result to satellite reentry [15, 41]. Detailed analysis and/or computation of drag force impact on Earth-orbiting satellites can be found in our earlier works [32, 33]. In the present circumstance, the earth orbit is not the reference, but the Martian orbit is expected to slowly circularize and MOM may crash onto Mars after a few tens of years.

6.4.1. *Computation of satellite orbital decay due to atmospheric drag*

To compute the actual earth orbital decay during the first month, the *NRLMSISE-00* empirical atmospheric model into our drag model has been incorporated. We have analyzed the drag effects due to space environmental perturbations on the trajectory of Mangalyaan satellite. We have assumed the satellite having a projected surface area of 5.04 m^2, a mass of 1337 kg and orbiting the Earth at an initial injected elliptical orbit of radius 250 km perigee by 23,500 km. We have chosen a spherical polar coordinate system (r, $\emptyset$) having origin $r = 0$ at the center of the Earth and assume that the satellite always remains in the same plane (i.e., $\theta =$ constant). The effects of the drag force have been computed from two sets of equations. The first set consists of four coupled differential equations represented by Eq. (10) [32, 33]:

$$\begin{aligned} \dot{\upsilon}_{\mathrm{r}} &= -\frac{\mathrm{GM_e}}{r^2} + r\dot{\phi}^2, \quad \dot{r} = \upsilon_r, \\ \ddot{\phi} &= -\frac{1}{2} r\rho\dot{\phi}^2 \frac{A_{\mathrm{s}} C_{\mathrm{d}}}{m_{\mathrm{s}}}, \quad \dot{\phi} = \upsilon_\phi / r \end{aligned} \tag{10}$$

where v_{r} and $v_{\emptyset}$ are the radial and tangential velocity components. G is the gravitational constant, M_{e} is the mass of the Earth, r is the instantaneous radius of the orbit, ρ is the atmospheric density, A_{s} is the omni-directional projected area of the satellite, m_{s} is the mass of the satellite and C_{d} the drag coefficient at an altitude of r. The four differential equations have been solved to obtain instantaneous positions and velocity components of the satellite in an orbit. To measure the decay of the orbital radius per orbit, we assume that the energy is constant per orbit. Incorporating the solution of orbit semimajor axis decay rate for near-circular orbit [15], we compute drag impact on the model Mangalyaan satellite under varying space weather condition while tracking its position and time by the tangential component parameters. Computations have been done using differential equation of changes in the mean radius of the satellite orbit per revolution (MRPR) [12, 22, 31, 34].

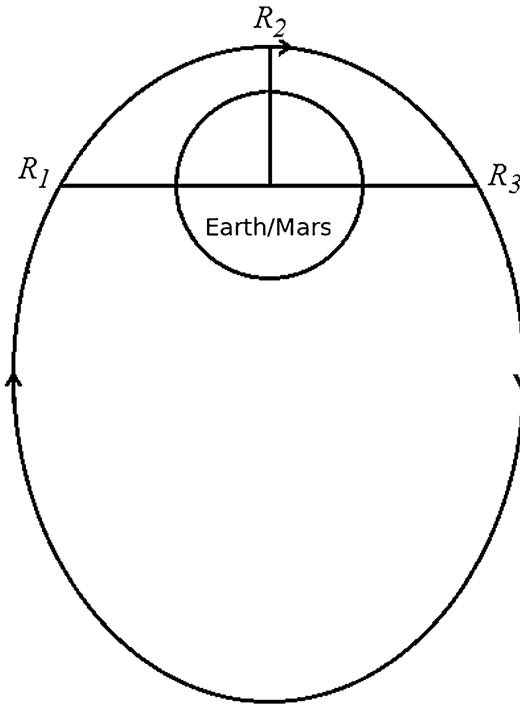


Fig. 5 Region of the orbit R_1 to R_3 where the drag effects have been considered, while the spacecraft is in Earth and Martian orbits

$$\frac{\mathrm{d}r}{\mathrm{d}t} = -\rho \frac{A_{\mathrm{s}} C_{\mathrm{d}}}{m_{\mathrm{s}}} \sqrt{\mathrm{GMr}} \tag{11}$$

6.4.2. *Heliocentric phase trajectory*

We have assumed a Hohmann-transfer heliocentric trajectory from Earth to Mars [15]. In heliocentric motion, the formulation of the above equations including the perigee and apogee velocity is with respect to the sun. The required perihelion velocity v_{p} and apoapsis velocity v_{a} at transfer orbit are given by the Eqs. (2) and (3).

6.4.3. *Earth and Martian atmospheric density profile*

The biggest challenge associated with computation of atmospheric drag force is the density profile of the atmosphere because it is not precisely known at any given instant. However, there are empirical atmospheric models with good approximation, such as the one used in this study. The *NRLMSISE-00* model gives outputs of altitude profiles of temperature, number densities of Helium, Oxygen (and its molecule), Nitrogen (and its molecule), Argon and Hydrogen, in equilibrium at the temperature, total mass (atmosphere) density and the number density of a high-altitude ‘anomalous’ oxygen component of total mass density that is not in thermal equilibrium at the temperature $T(z)$. This model is mainly developed for the Earth’s atmosphere. Although Mars is similar to Earth in many ways, but their atmospheric constituents and/or conditions are not the same. For instance, the percentage composition of carbon dioxide and nitrogen in Martian atmosphere is about 93 and 2.7 %, while that of Earth is less than 1 and 78 %, respectively. In this analysis, without strict consideration for individual constituent of Martian atmosphere, we assume that the total mass density of Mars atmosphere is only one percent (1 %) that of Earth’s atmosphere. Again, we assume the geometry of the ellipse orbit as shown in Fig. 5 for analysis and computation of drag around and during satellite perigee passage in Earth and Martian orbits. We neglect drag effect around and during apogee passage and compute density (and drag) at an average distance (R_m) of

$$R_m = \left(R_1^2 + R_2^2 + R_3^2\right)^{1/2}.$$

Thus, the effect of drag is assumed to be significant only during the period when the satellite moves from R_1 to R_3.

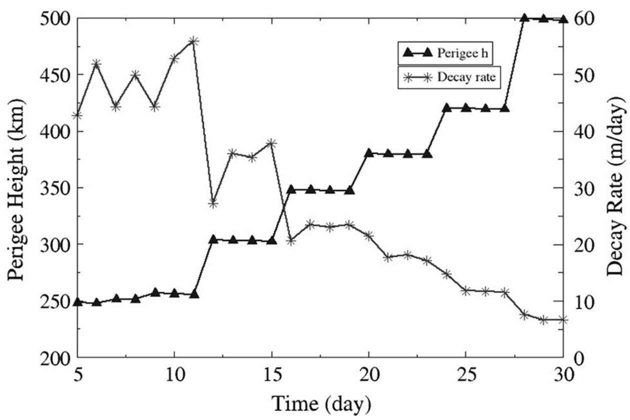


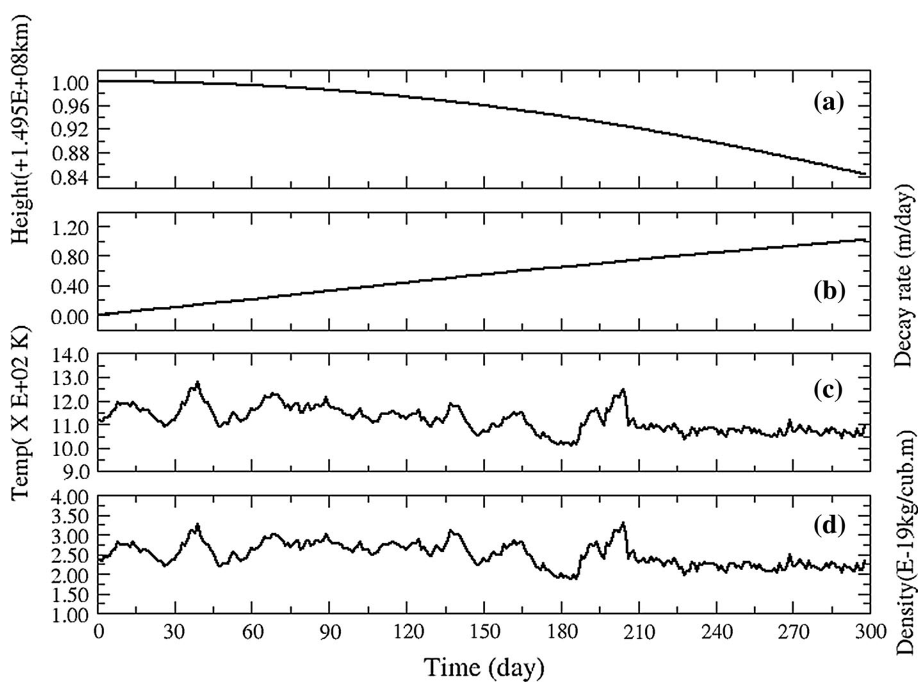
Fig. 6 Maneuvered orbits of Mangalyaan Mars orbiter (*Black curve* with *triangles*) with corresponding decay rate (*gray curve* with *stars*). *Triangles* correspond to number of days the MOM stayed before the orbit boost

7. Results and discussion

Mangalyaan altitude at maneuvered orbits and the corresponding computed decay rate are shown in Fig. 6. We find that the spacecraft experienced orbital decay rate of about 47.34, 47.14, 51.01, 34.12, 22.67, 18.62, 12.42 and 6.87 m/day at respective maneuvered orbits of 248.4, 252.0, 257.0, 304, 348, 380, 420 and 500 km. The triangles at specified heights indicate the number of days the spacecraft stayed in corresponding orbit before velocity boost or orbit-rising. Wherever perigee height for each orbit boost/rise (maneuvered) is not explicitly provided, reasonable values were assumed.

The rate at which the spacecraft decayed decreased with increased altitude (subsequent to orbit-rising). This is expected since the density and drag force are reduced with increase in height from the earth depending on value of solar and geomagnetic index. Although orbit-rising reduces

Fig. 7 Mangalyaan (**a**) mean altitude, (**b**) orbit decay rate, (**c**) thermospheric temperature and (**d**) density during heliocentric trajectory



drag force on the spacecraft, we have observed that severe space weather condition, especially during high geomagnetic activity, increased the drag force sporadically on the system. Orbit decay rate increases significantly on the ninth and eleventh days (after launch date), due to high solar and/or geomagnetic activity. Upper atmospheric heating and associated density fluctuations are largely due to solar EUV. However, geomagnetic field-induced heating is important during short interval of geomagnetic disturbances and/or storms [32–34]. We find that when perigee height increased by a total sum of about 250 km, it decayed by a total of about 720 m due to atmospheric drag force within about 28 days in the Earth's orbit during geocentric trajectory.

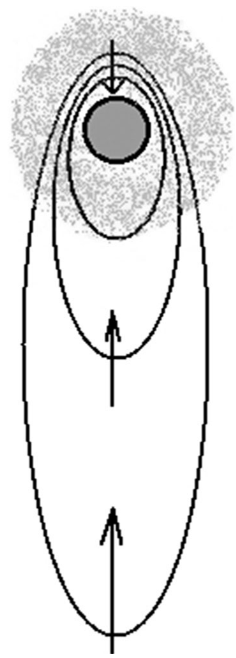


Fig. 8 Schematic diagram of Orbital decay effect (circularization) on Mangalyaan in Martian orbit

Figure 7(a)–7(d) show computed (a) mean altitude of Mangalyaan, (b) rate of its orbit decay, (c) thermospheric density and (d) thermospheric temperature during the heliocentric trajectory for 10 months. We have assumed an elliptical orbit with mean radius (altitude) of 149,502,370 km during Earth–Mars (heliocentric) transfer. Typically, thermospheric temperature varies between 1050 K and approximately 1350 K and density varies between 2.0×10^{-19} and 3.5×10^{-19} kg/m^3. The total orbital decay is about 157.6 m. These calculations and/or values are based on the indices of solar and geomagnetic activity in the intervening period as inputs. Atmospheric drag and subsequent decay that may be experienced by Mangalyaan during heliocentric trajectory (300 days) is small compared to geocentric (28 days) trajectory scenario (up to a factor of four). Although quite nominal, but the implication of the seemingly small amount of decay is that the planned orbital parameters (such as the Periapsis and Apoapsis) at Mars orbit insertion may be influenced. This unforeseen change must be corrected before Mars arrival Phase. On the other hand, if the expected orbital parameters and hence the retro velocity requirement are not critical (stringent requirement), fluctuation of this magnitude (157.6 m) may not influence the craft's safe Mars orbit insertion.

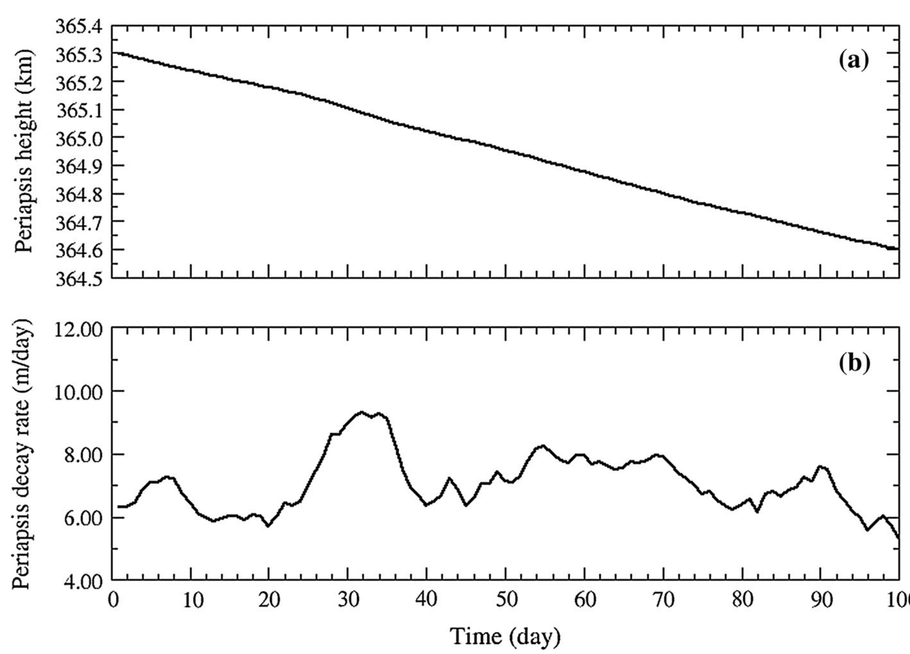


Fig. 9 Mangalyaan (**a**) mean periapsis height and (**b**) decay rate during first 100 days of Martian trajectory

Figure 8 shows the computed decay profile of the spacecraft after its insertion into Martian orbit. The Martian atmosphere causes a drag only on a small part of the planned 365 km × 80,000 km elliptical orbit. Since the density in Mars atmosphere is only about 1 % of that of the Earth, drag effect is expected to be very minimal. However, this becomes important over a long term. For a very small decay of apoapsis, the periapsis would decay by a large amount and thus, eventually the orbit would be circularized. Under such scenario, the orbit decays faster and would theoretically crash into Martian atmosphere. Figure 9(a) and 9(b) show (a) the mean periapsis height and (b) decay rate of Mangalyaan during 100 days trajectory after Mars orbit insertion. Our result shows a total decay of about 700.8 m and a decay rate of up to 9 m/day in areocentric orbit. Our computation is based on the actual observed indices of solar and geomagnetic activity as inputs between September 24, 2014 and December 31, 2014 (after MOI), and assumed atmospheric density profile of Mars. Calculation (and/or model) of Mars atmosphere is based on Mars–Earth atmosphere (about 1 %) and temperature ratio using the *NRLMSISE-00* empirical model. We are now working on developing approximate Martian atmosphere density model for more accurate calculation and/or prediction. This will be submitted elsewhere.

8. Conclusions

In this study, natural perturbing forces associated with Mars (and/or interplanetary) mission during geocentric, heliocentric and Martian trajectories have been identified and their possible effects analyzed. These forces are mainly planetary and/or solar-forcing-induced. Earth's oblateness (J_2), Third body (solar and lunar), solar radiation pressure, solar energetic radiation environment and atmospheric drag forces can influence the trajectory of the mission. We also have modeled the impact of atmospheric drag force on Mangalyaan Mars orbiter mission as a function of appropriate space environmental parameters during about 28 days of geocentric trajectory, 300 days of heliocentric, and 100 days of areocentric trajectory. We have found that for a total perigee height boost of about 250 km (in Earth orbit), the cumulative orbit decay can be approximately 720 m. Our computation suggests that approximate altitude variation was up to 158 m with respect to the sun during 300 days of interplanetary journey toward Mars. After MOI, the total decay experienced by the spacecraft would be about 701 m with decay rate of up to 9 m/day during 100 days of Martian trajectory based on Mars–Earth atmosphere density ratio. In our analysis, we find that in all the three phases, only minor corrections are required. With actual density model of the Martian atmosphere, we are now in a position to compute drag over the whole solar cycle and the results will be published in another paper.

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